

TPMS Explorer: an interactive, gate-verified open ecosystem for triply periodic minimal surface design, generative implicit fields, multifunctional physics simulation, and biophysical tissue-ingrowth dynamics

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Abstract

We present TPMS Explorer v8.0, a browser-based, zero-backend ecosystem for parametric design, physics simulation, and additive-manufacturing preparation of triply periodic minimal surface (TPMS) architected materials. The platform couples a watertight Surface Nets meshing pipeline with forty-two continuously executed CI gates ($> 1,000$ assertions under strict per-line accounting, each gate carrying a minimum-assertion guard) that re-execute the exact code claim shipped to the user. The surface library spans 20 level-set families — the eight canonical topologies plus C2 extensions (O,C-TO; Karcher; Fischer-Koch S/Y/C(S)/C(Y); G'; D'; Double P/D/G; Complementary D) transcribed from the MiniSurf reference implementation, with four-way formula parity (TypeScript / Python / MATLAB / GPU IR) and a public geometric benchmark matrix. Version 7.0 adds five research-grade modules: (i) a sinusoidal-activation implicit neural field (SIREN) with an exactly periodic Fourier-feature input and an 8-dimensional latent space over five distilled TPMS experts; (ii) multiaxial anisotropic yield envelopes (Hill-48, Tsai-Wu, Gurson, Drucker-Prager) with a unified star-convex ray-distance formulation and an interactive 3D viewer; (iii) a Bloch-Floquet phononic band-structure solver built on Born-von Kármán lattice dynamics with realified two-pass deflate-Lanczos eigensolution and long-wave dynamical-matrix calibration; (iv) a reaction-diffusion tissue-ingrowth engine coupling quasi-steady oxygen transport, hypoxia-gated osteoblast proliferation, and mineral deposition over a 28-day horizon; and (v) a level-set topology-optimization loop with Hamilton-Jacobi Godunov evolution driven by elastic and flow sensitivities under a volume constraint. Post-release hardening adds (vi) a deterministic exact-porosity solver — analytic-integration root finding with mesh-measured secant validation, reaching 0.26 pp deviation at resolution 96 — and (vii) constructive global orientation propagation that reduces misoriented STL edges to zero by construction, replacing a per-triangle normal-voting scheme that degraded winding consistency five-fold. Version 8.0 adds (viii) an agentic verification layer: a five-operation deterministic tool schema consumed through LLM function calling under a per-slot clamping interceptor (enumeration, range, unknown-property, and path-traversal rejection), and a closed-loop design-verification driver in which the LLM selects repairs only from a bounded strategy menu while patch application, execution, and acceptance remain deterministic — including structured unreachable declarations for representation-limited parameter combinations. All simulation modules execute client-side with zero dependencies beyond Three.js; the agent layer runs under Node.js and exports to industrial formats (STL, 3MF, GLB, Abaqus INP, OpenFOAM polyMesh, G-code), guarded throughout by behavior-level audit gates whose calibration traps are documented.

Keywords: TPMS, implicit neural representation, SIREN, yield surface, phononic bandgap, tissue engineering, level-set topology optimization, LLM tool calling, continuous integration

1. Impact overview

Triplically periodic minimal surfaces — first catalogued systematically by Schoen Schoen (1970) — are today a mainstay of architected-materials design and bone-scaffold engineering, where matching a target porosity while keeping the lattice manufacturable is the central numerical problem. TPMS Explorer turns a browser tab into a research bench for architected materials. The five v7.0 modules extend the platform from “geometry plus linear physics surrogates” to generative design, structural-failure envelopes, wave dynamics, biophysical growth, and closed-loop shape optimization — each guarded by an audit gate that executes the same code paths the user sees (Table 1). Version 8.0 completes the loop on the automation side: an agen-

tic verification layer turns natural-language design intents into schema-clamped tool calls and drives verification-repair cycles to convergence, with every acceptance decision still taken by deterministic code (Section 3.8).

2. Necessary resources

2.1. Hardware

Any 2020-class laptop. WebGPU is optional (acceleration only); all v7.0 modules run on the CPU path.

2.2. Software

Modern Chromium/Firefox/Safari; Node.js ≥ 20 for the CI suite; TypeScript 6, Vite 8, Three.js 0.185 (only runtime depen-

dency).

3. Method design

3.1. SIREN implicit field with exact periodicity

A 4-layer sinusoidal network in the spirit of SIREN Sitzmann (2020), $F_\theta(\mathbf{x}) = W_3 \sin(W_2 \sin(W_1 \sin(W_0 \mathbf{u}))) + \mathbf{b}_3$ consumes the Fourier feature $\mathbf{u} = [\sin x, \cos x, \sin y, \cos y, \sin z, \cos z]$. Because every coordinate enters only through $\sin / \cos$, F_θ is exactly 2π -periodic (measured deviation 2.2×10^{-15}), which makes cross-period water-tightness a property of the architecture rather than of the fit. Five experts (Gyroid, Diamond, Lidenoid, Schwarz P, fractal trabecular) are distilled offline from the platform’s single semantic source of TPMS fields with Adam (RMSE 0.69%–8.04% at 16^3 , hardest for the multi-scale teacher). A latent code $\mathbf{z} \in \mathbb{R}^8$ selects experts through $m = \text{softmax}(-\|\mathbf{z} - \mathbf{A}_i\|^2 / 2\sigma^2)$ with ± 3 Walsh anchor codes (pairwise Hamming distance 4, cross weight e^{-72}). Lipschitz continuity is bounded by $\prod_i \|W_i\|_2$ (power iteration) and cross-checked against sampled gradient suprema.

3.2. Multiaxial yield envelopes

All four criteria are expressed in a unified *ray-distance* form: the distance $r(\hat{\mathbf{n}})$ from the origin to the envelope along a unit direction of principal-stress space. Hill-48 Hill (1948) (with a hydrostatic cap), capped Drucker–Prager, and Gurson close to star-convex bodies containing the origin, hence r is single-valued and the safety factor is exactly $r(\hat{\mathbf{n}})/\|\sigma_0\|$. Tsai–Wu admits a closed-form quadratic root. Gurson’s hydrostatic pole Gurson (1977) $(2\sigma_0/3) \text{arccosh}((1+q^2)/2q)$ with q equal to the porosity provides a strong analytic anchor. Scaffold-level parameters are derived from the same Gibson–Ashby constants Gibson (1997) used by the platform’s energy-absorption module.

3.3. Bloch–Floquet phononic bands

The unit cell is discretized into voxel masses connected by 26-neighbourhood axial springs; the Bloch–Floquet treatment follows the standard phononic-crystal formalism Hussein (2014). Bloch phases assemble a complex Hermitian $K(\mathbf{k})$, realified as $\begin{pmatrix} A & -B \\ B & A \end{pmatrix}$ so that all eigensolution machinery is real. Two-pass *deflate-Lanczos* (full reorthogonalization; the second run is deflated against the first run’s Ritz vectors) recovers both copies of every complex eigenvalue, preserving degeneracies. Spring stiffness κ is calibrated by exact inversion of the long-wave dynamical matrix $\omega_{LA}^2 = \kappa h^2 k^2 S_x / (mN^3)$, $S_x = \sum dx^4/d^2$ — affine homogenization is explicitly *not* the long-wave modulus of a central-force network (measured ratio 0.53, matching the analytic 0.54). Under periodic boundary conditions rotations are not admissible Bloch modes, so the Γ point carries exactly three translational zero modes.

3.4. Tissue ingrowth reaction–diffusion

Oxygen obeys a quasi-steady diffusion–Michaelis–Menten balance solved by 7-point Jacobi iteration with no-flux mirrors at solid interfaces and Dirichlet C_0 on the perfused boundary layer; osteoblasts follow hypoxia-gated logistic growth with explicit daily updates; mineral accumulates where oxygen exceeds a threshold. Operator splitting is justified by $\tau_{O_2} = L^2/D \approx 0.05$ days versus a 28-day proliferation horizon. $V_{\max}$ is calibrated to a literature oxygen penetration depth $L_{\text{pen}} = \sqrt{2DC_0/V_{\max}} \approx 0.4$ mm. The gate verifies the discrete mass balance (boundary-layer flux = interior consumption, residual 0.000%) and hypoxia gating on a dense-versus-porous pair.

3.5. Level-set topology optimization

The TPMS signed field evolves under $\partial\Phi/\partial t + V_n|\nabla\Phi| = 0$ (Godunov upwinding) with V_n assembled from elastic von-Mises sensitivities (fully-stressed) and Darcy pressure-gradient sensitivities, zero-meant over *interface* voxels, normalized, and constrained by a volume Lagrange bias. Classical compliance under a unit service load is proportional to $1/\text{reaction}$ of the displacement-controlled solve — the $\frac{1}{2}F\delta$ reading is a softening artefact under displacement control. Islands are removed by largest-component BFS, Φ is reinitialized by an exact 3D Euclidean distance transform, and the final field re-enters the platform’s Surface Nets pipeline through the precomputed-field injection contract, guaranteeing water-tight extraction.

3.6. Exact porosity solver: analytic-integration root finding

The default porosity path couples the requested value to a voxel-rank bisection whose caliber differs from the delivered triangle mesh; measured deviations reached 24 pp for higher-harmonic families at coarse resolutions. Following the analytic-integration methodology of RegionTPMS Karakoç (2021), the agent CLI now (i) locates iso^* on the *analytic* field by bisection over a deterministic Monte-Carlo integration of $P(F < iso)$ in the unit cell (LCG-seeded, bit-reproducible), then (ii) builds once with iso^* (bypassing the internal bisection) and applies a single secant correction along the analytic slope to absorb the mesh-versus-analytic volume gap, reverting to the direct build if the corrected attempt is worse. Measured deviations: 0.26 pp (Diamond, 65% target, resolution 96) and 1.3–4.3 pp across cubic-symmetric families at resolution 48; higher-harmonic families at resolution 48 remain representation-limited (9–23 pp) and are reported honestly with an upgrade hint. A cube + Diamond + resolution-64 combination with non-manifold edges (nm = 740, down from 1940 under the legacy path) is fail-closed by the water-tightness gate.

3.7. Global orientation propagation

The binary and multi-solid STL writers previously flipped each triangle independently by comparing geometric normals against the vertex normal field; because that field is unreliable at cap fans and boundary skins, independent voting fragmented a locally consistent winding (misoriented edges 4536 $\rightarrow$ 23571 after flipping at resolution 96). The writers now call a constructive orientation pass: register every undirected edge with its two

incident triangles, two-color the triangle adjacency graph by BFS (same-direction neighbours get opposite colors), apply the parity, and flip the whole mesh once if the divergence-theorem volume is negative. For the closed manifolds the pipeline guarantees (open = non-manifold = degenerate = 0), misoriented edges are zero by construction — verified byte-level on the exported STL (823,500 edges at resolution 96).

3.8. Agentic closed-loop verification (v8.0)

The agent layer follows one iron rule: the LLM only fills schema-bounded slots, and every numeric value that reaches the geometry pipeline is clamped or rejected by a deterministic interceptor — enumeration membership, numeric ranges, unknown-property rejection, and single-segment path patterns (anti path-traversal) are enforced per slot, so adversarial or malformed model output can neither inject code nor escape the parameter domain regardless of model behavior. Five operations are registered in the tool schema: `list`, `estimate`, `mesh` (direct watertight build), `scenario` (end-to-end scaffold delivery: STL + Abaqus INP + verification report), and `design-verify` (the closed-loop entry). A natural-language agent maps bilingual instructions — with alias resolution such as Schwarz D → diamond — to tool calls through LLM function calling over three interchangeable providers: a local Ollama endpoint, any OpenAI-compatible endpoint, and an offline Mock provider whose real-execution path is guard-railed behind an explicit opt-in so that closed-loop regression can run without network access.

The design-verify driver closes the loop: propose → verify (water-tightness, porosity-deviation, and physical-range checks) → on failure, the LLM selects the next repair *only* from a bounded strategy menu (surface-family swap, period reduction, resolution escalation ≤ 128, container or mode change, per-slot parameter fixes) → deterministic patch application and re-run. Repairs inside the existing verification ladder (automatic resolution escalation, secant correction) remain deterministic; the driver handles ladder-external repairs, and for representation-limited combinations (e.g. the cylinder + diamond deep-water region identified by the benchmark matrix) it emits a structured *unreachability declaration* instead of looping. Acceptance rests on three layers: a 34-instruction bilingual regression against a production LLM (34/34; all adversarial samples rejected or self-clamped); an offline mock-driven closed-loop self-test covering parameter-fix convergence, out-of-ladder resolution repair, period reduction, unreachable declaration, interceptor rejection, and round exhaustion with the original design file bit-identical; and a 87-assertion schema↔CLI cross-check gate that pins enumeration parity across tools, a static sentinel between the schema type enumeration and the driver’s repair menu, and an offline end-to-end execution of the bridge. The LLM regression itself is key-gated and runs outside CI; correctness of the layer does not depend on it because the interceptor is deterministic.

4. Quality control

Forty-two gates run via `npm run test:all` on three platforms. Every gate loads the shipped TypeScript through a bundler

and asserts behavior-level invariants (no mocked math), and every gate now carries a *minimum-assertion guard*: a probe experiment showed that an auditor whose assertions are all neutralized still exits green under the historical `if (fail) exit` idiom, so each gate additionally fails when its executed-assertion count drops below a measured baseline. Three agent-layer self-checks whose unregistered periods had incident histories of silent redness (a drifted fail-closed contract that went unnoticed for a day) were promoted into the CI scheduler so that the same discipline covers the automation layer itself. Four tautological assertions (two `|| true` disjunctions, a self-multiplying metric check, and a dead variable that was computed but never asserted) were exposed by a dedicated audit and either removed or rewritten as real checks; one of them had been masking a wrong closed-form slope in the Poincaré extension for its entire lifetime. Gate calibration is treated as a first-class activity: documented false-red episodes (a $\sqrt{2}$ scaling slip in the auditor von-Mises conversion; a Gram–Schmidt normalization threshold that silently skipped 10^{-10} -scale columns; a cross-implementation static sentinel left unsynced after a formula change) are recorded and their fixes covered by regression assertions.

5. Availability

5.1. Current code version

v8.0.0-agentic-loop (MIT license).

5.2. Operating requirements

See Necessary resources; no backend, no telemetry, no build-time secrets.

5.3. Installation

Open `docs/platform/index.html` (static build) or run `npm install` and `npm run dev` in `tpms/tpms-platform`. CI: `npm run test:all`. A deterministic agent CLI (`tpms/agent/tpms`) exposes `list`, `estimate`, `mesh`, `verify`, `solve`, and `scenario` subcommands with JSON output for scripting; `llm-agent.mjs` adds natural-language tool calling and `tpms-driver.mjs` closed-loop design verification (LLM API keys are supplied via environment variables and never committed).

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Table 1: v7.0/v8.0 modules and their CI gates.

Stage	Module	Verified invariants
I	SIREN implicit field	exact 2π periodicity (2.2×10^{-1} distillation RMSE per expert; Lipschitz product bound vs. sampled gradients; latent-path Lipschitz; watertight meshes at anchors and manifolds at midpoints
II	Yield envelopes	uniaxial analytic anchors; midpoint convexity ($\times 200$ per criterion); isotropic permutation symmetry; Gurson hydrostatic pole; closed mesh volume by divergence theorem; safety factor of a half-scaled structure = 2
III	Phononic bands	Γ -point triple translational zero modes; sound-speed self-consistency (gyroid 2.2%); per- k branch ordering; time-reversal symmetry $\omega(\mathbf{k}) = \omega(-\mathbf{k})$; diatomic-chain analytic gap; island-count guard
IV	Tissue ingrowth	mass balance = boundary-layer flux – interior consumption (0.000%); core-oxygen transport differential equation; $\rho \leq \rho_{\max}$; hypoxia gating; mineral monotonicity; bitwise determinism
V	Level-set optimization	unit-load compliance $\propto 1/\text{reaction}$; monotone under 0.5% tolerance; volume Lagrange constraint $\pm 10\%$; permeability gain $\times 1.50$; water-tight contraction (open edges = 0); reinitialization $ \nabla\Phi \approx 1$
VI	Agentic verification (v8.0)	schema $\leftrightarrow$ CLI cross-checks (enum/range/path parity, incl. sentinel vs. driver repair menu); interceptor clamping incl. adversarial and malformed calls; mock-driver closed-loop convergence on injected defects with the original design for bit-identical